

RQ3 - Temporal Development in GlobalGeoTree

Changes in tree observation volume, source composition, and geographic coverage from 2015 to 2024

Research Question

RQ3 - Temporal Development

How has the number and distribution of tree observations changed from **2015 to 2024**?

This report answers four subquestions:

- 1. Has the number of observations increased over time?
- 2. Which years show the strongest growth or decline?
- 3. Did geographic coverage expand over time?
- 4. Are temporal patterns mainly driven by specific sources?

The key idea is that time trends in GlobalGeoTree may reflect both real changes in observation coverage and the growth of data platforms such as iNaturalist and Pl@ntNet.

Data Loading

Measure	Value
Observation file	GlobalGeoTree.csv
Period analysed	2015-2024
Rows analysed	6,263,345
Distinct sources	2,493
Distinct source groups	6
Distinct countries	221
Distinct species	21,001
Coordinate columns available	Yes

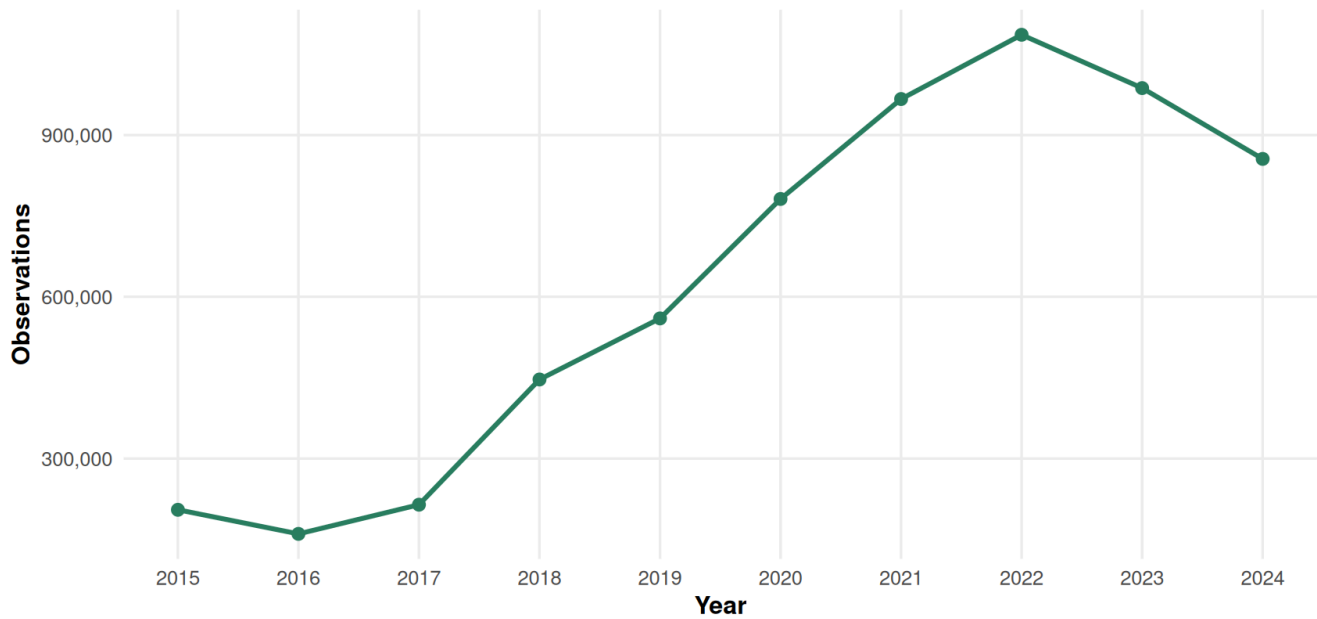
This report uses 6,263,345 observations from 2015 to 2024.

Observation Growth Over Time

Between 2015 and 2024, observations changed from **205,055** to **855,708**, an overall change of **317.3%**.

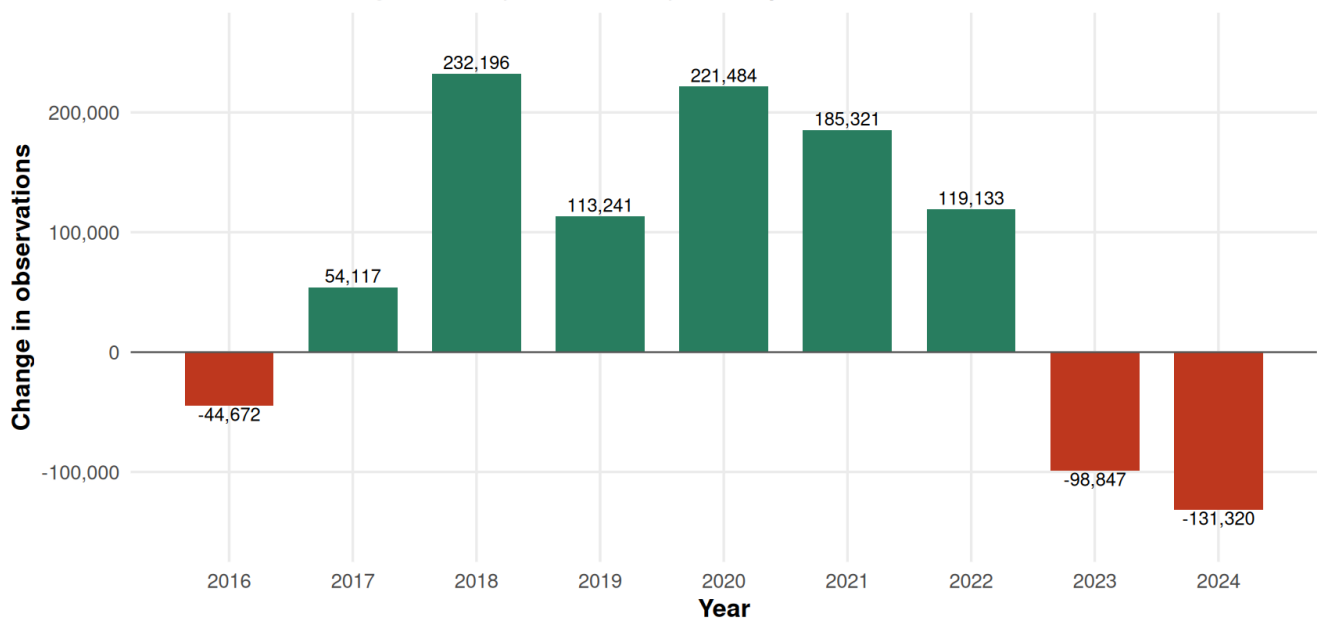
Number of tree observations by year

Observation volume in GlobalGeoTree from 2015 to 2024.



Year-to-year change in observations

Positive bars indicate growth compared with the previous year.

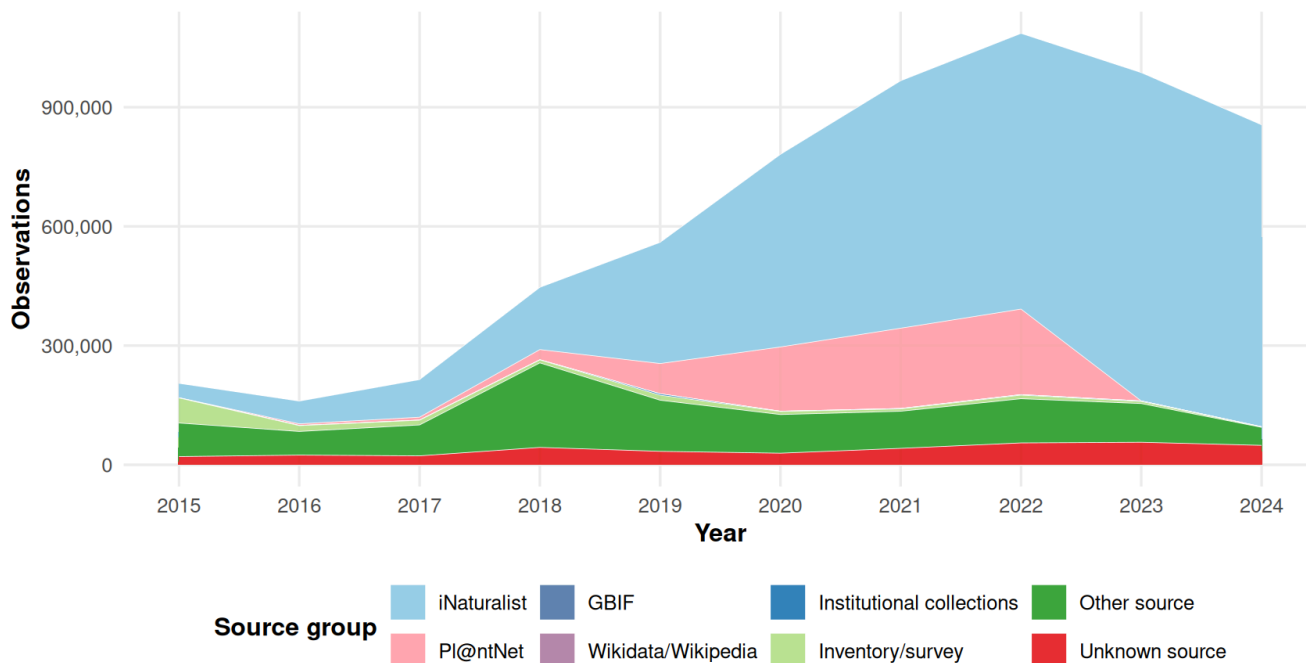


Interpretation. The strongest growth occurs in **2018**, with **232,196** more observations than the previous year. The strongest decline occurs in **2024**, with **-131,320** fewer observations than the previous year. These changes should be interpreted together with source composition, because platform growth can create strong temporal trends.

Are Temporal Patterns Driven by Specific Sources?

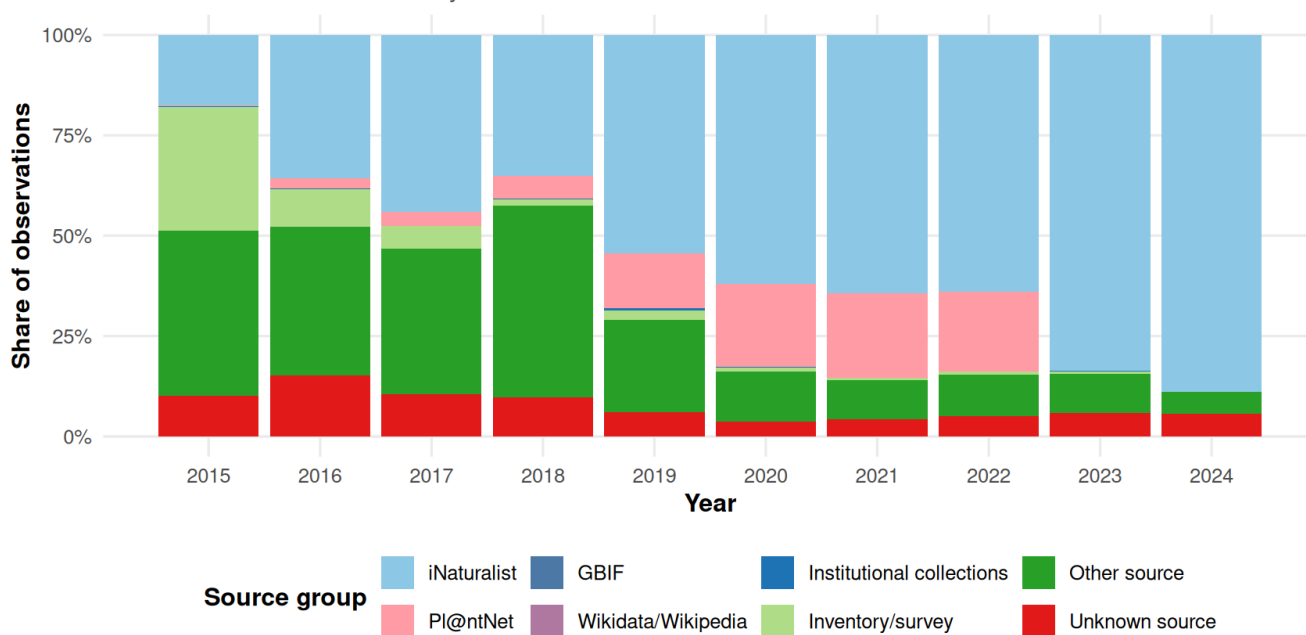
Observation volume by source group and year

A rising source group can drive the apparent growth of the full dataset.



Source composition by year

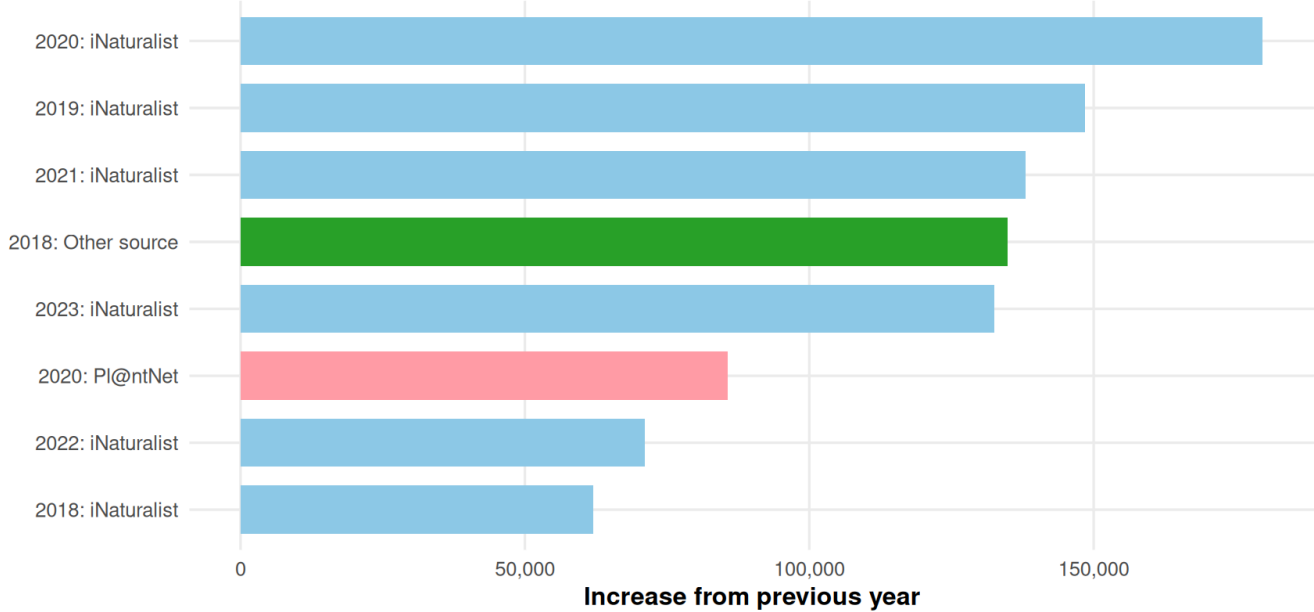
Bars sum to 100% within each year.



The most recent year may be incomplete depending on the data export date.

Largest source-specific yearly increases

This identifies which source groups contributed most to single-year growth.

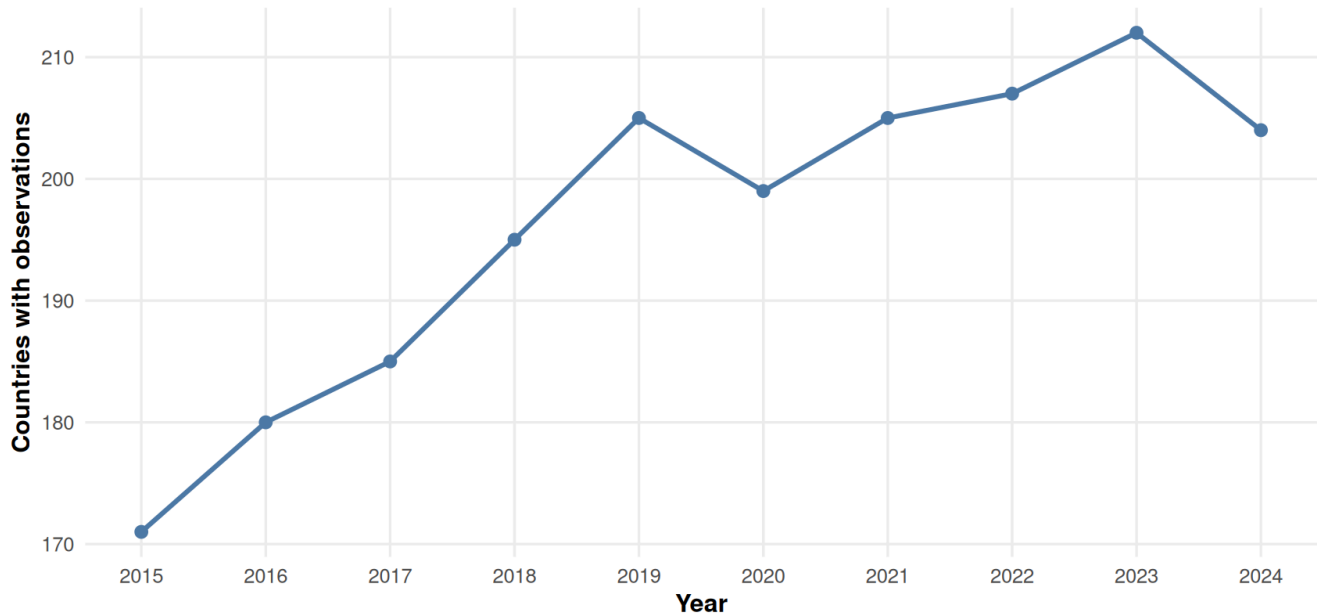


Interpretation. If most growth is concentrated in one source group, the temporal trend is source-driven. In that case, the line chart should not be interpreted as a neutral increase in global tree recording.

Geographic Coverage Over Time

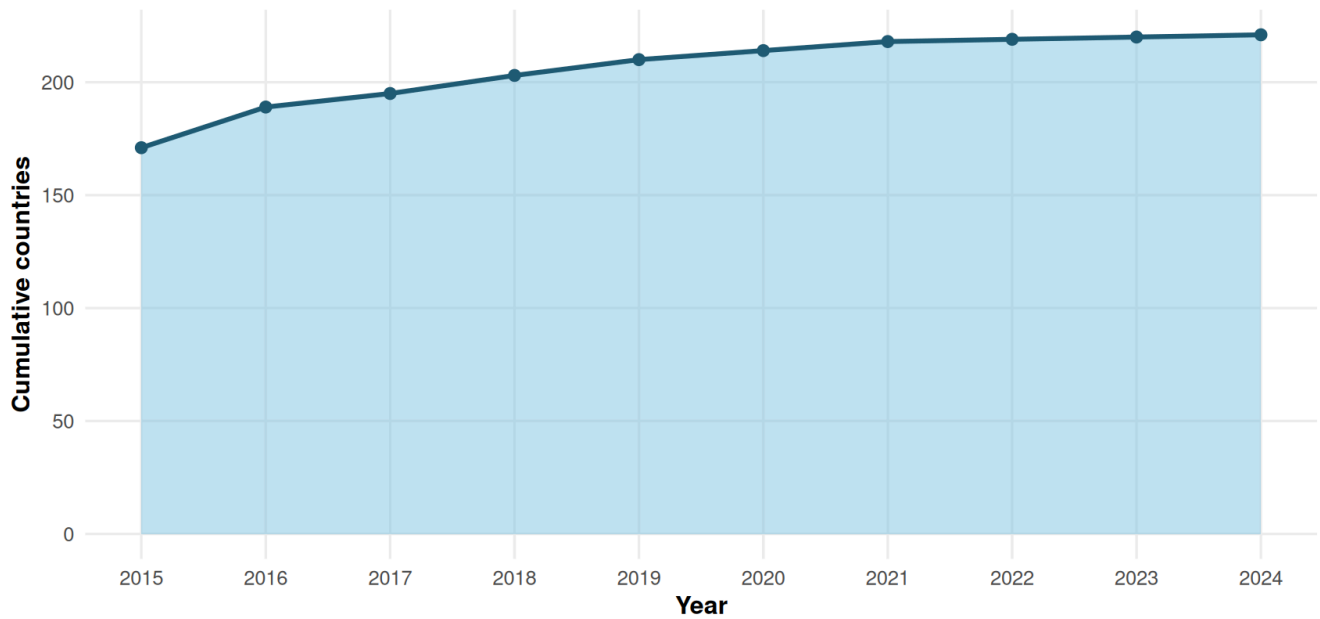
Countries represented each year

Annual country coverage shows whether observations became geographically broader.



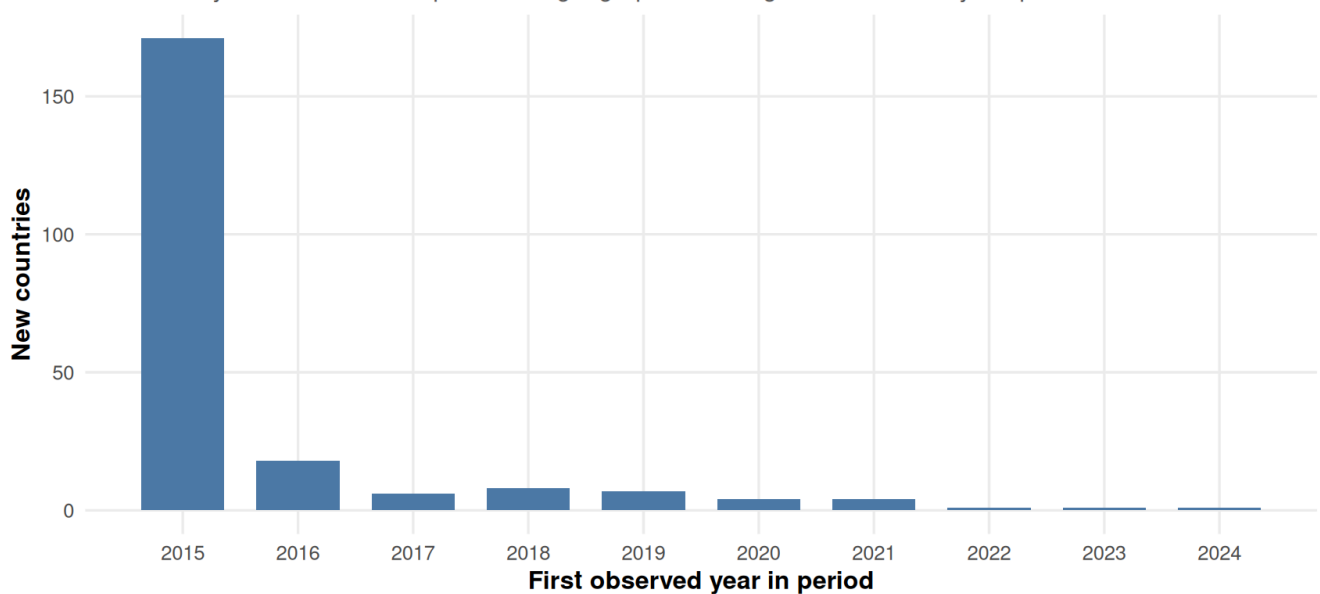
Cumulative country coverage over time

The curve rises when a country appears in the analysed period for the first time.



New countries entering the dataset by year

New country entries indicate expansion of geographic coverage within the analysed period.

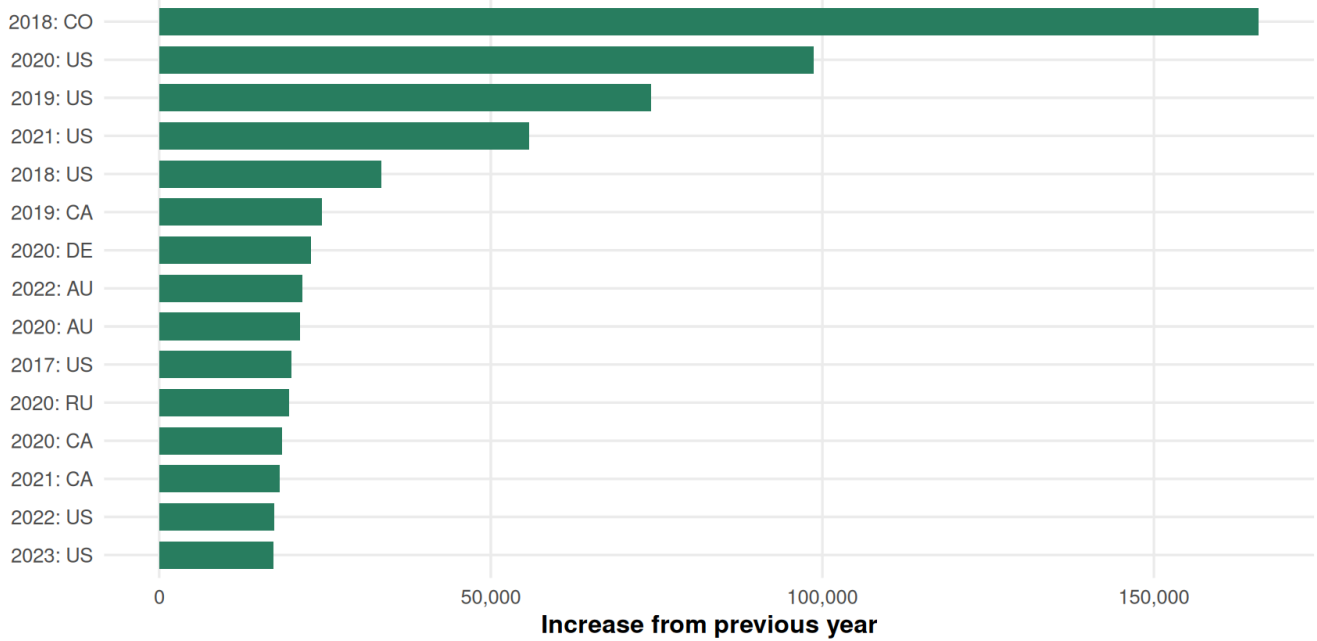


Interpretation. If the number of countries and cumulative country coverage increase over time, GlobalGeoTree is expanding geographically. If observation counts increase without much country expansion, growth may be concentrated in already well-covered countries.

Which Countries Drive Temporal Growth?

Largest country-specific yearly increases

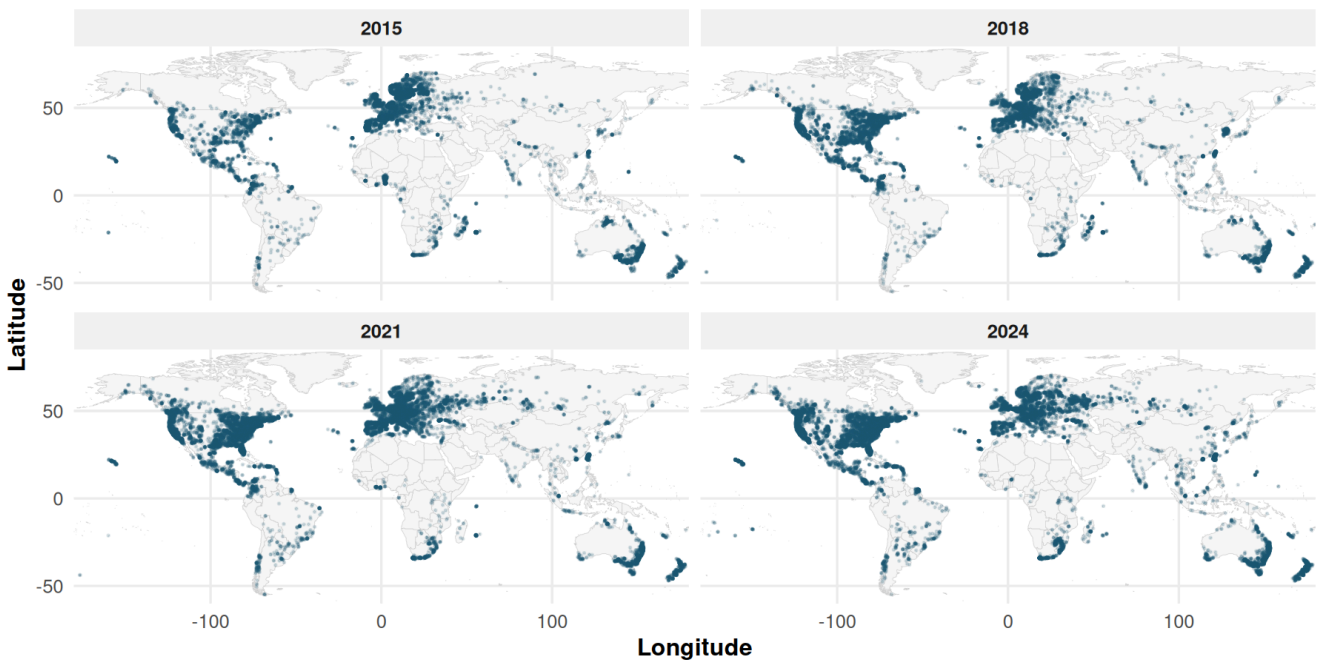
Temporal growth may be concentrated in specific countries rather than global.



Map Snapshots for Selected Years

Map snapshots of observations over time

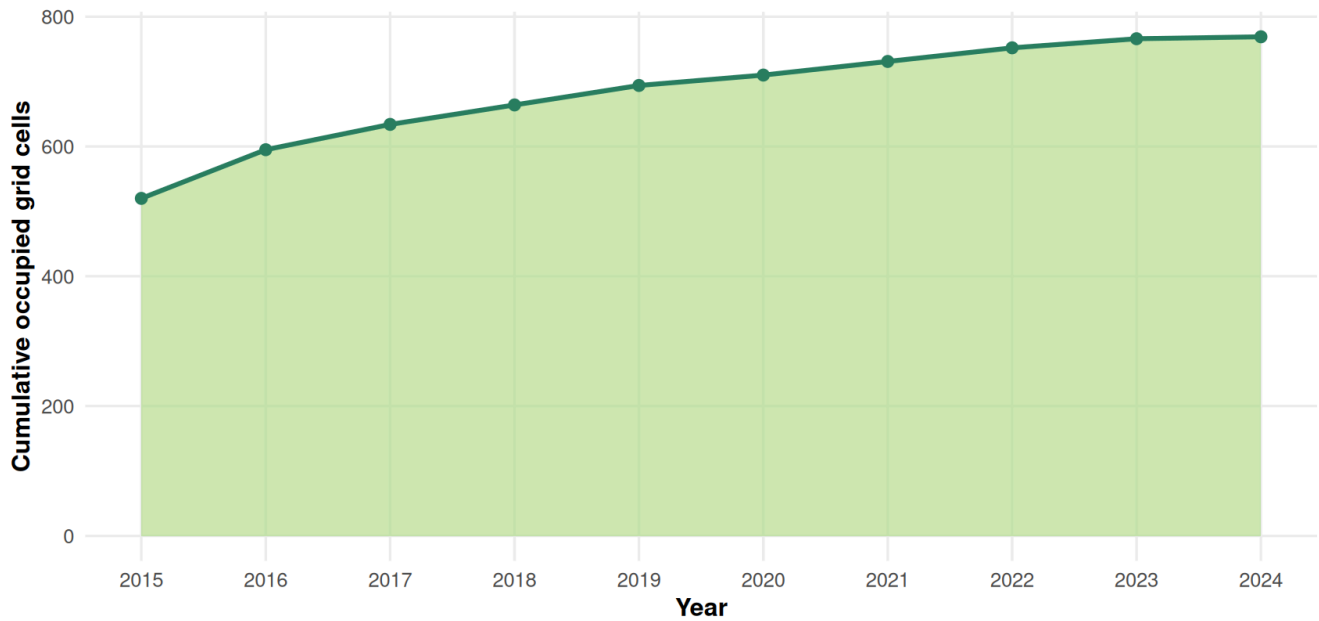
Selected years show whether records spread geographically or intensify in existing regions.



Points are sampled per year for readability. Install maps for country outlines.

Cumulative spatial grid coverage over time

A 5-degree latitude/longitude grid approximates whether spatial coverage expands.



Interpretation. Map snapshots and grid coverage distinguish two different processes: more observations in the same places versus expansion into new regions.

Overall Interpretation

From 2015 to 2024, temporal development in GlobalGeoTree should be interpreted as a combination of:

- changing observation volume,
- expansion or concentration of geographic coverage,
- country-specific growth,
- and source/platform growth.

If the strongest year-to-year increases are driven by citizen-science platforms or a small number of countries, the temporal trend is not purely biological. It is partly a record-production trend. Therefore, temporal analyses should report both total observations and source composition by year.